

Denoising of EBSD Data

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Noise in EBSD Data

As any experimental data orientation measurements are subject to errors

potential sources of measurement errors

- ▶ sample preparation
- ▶ distortion due to sample inclination
- ▶ exact pattern center
- ▶ heating of the sample
- ▶ limited resolution of the Kikuchi image
- ▶ limited resolution in Hough space
- ▶ limited resolution in orientation space
- ▶ pseudosymmetries

two classes of errors

- ▶ systematic errors
- ▶ random errors

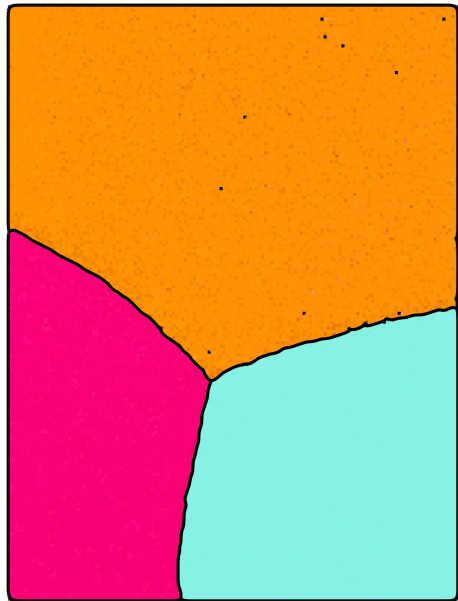
A Very Simple Example

- ▶ orientation map of undeformed pure aluminium
- ▶ compute grain boundaries
- ▶ colorize orientations according to misorientation angle and misorientation axis to the grain mean orientation (Thomsen et al.: Quaternion-based disorientation coloring of orientation maps, 2017)
- ▶ the kernel average misorientation
- ▶ the gradient in X direction
- ▶ “remove” noise
- ▶ KAM of the “noisefree” data



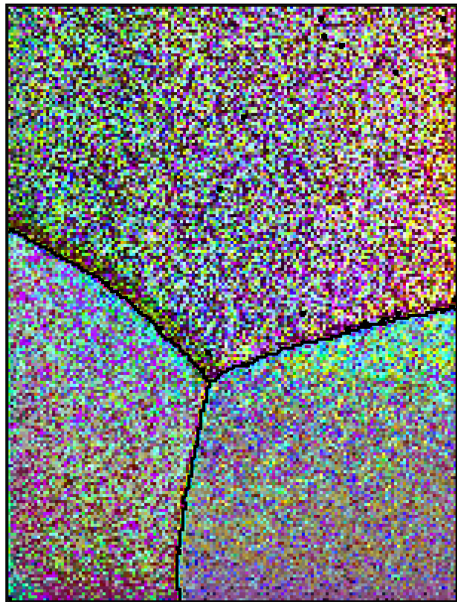
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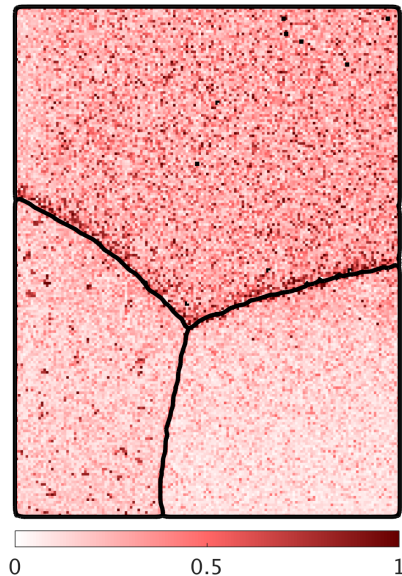
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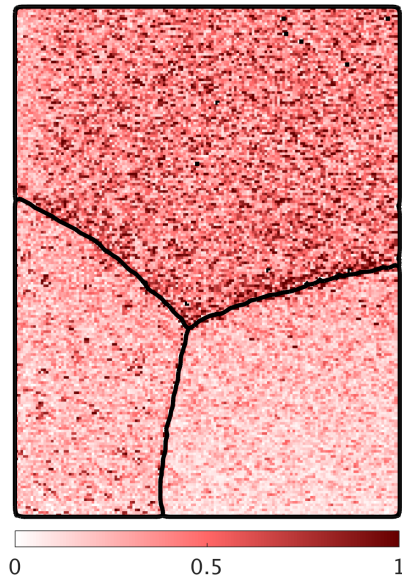
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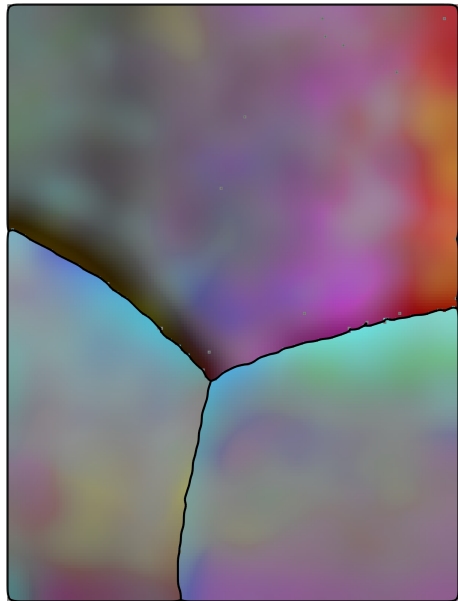
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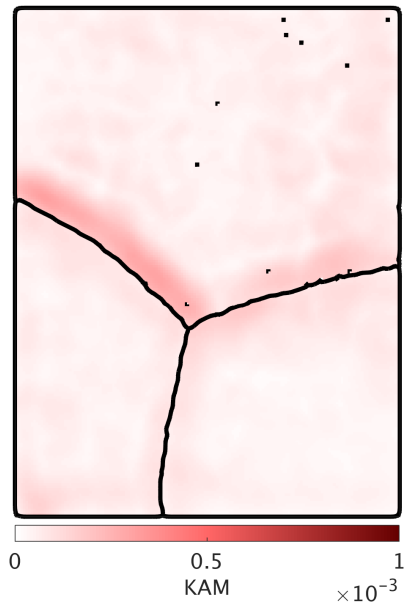
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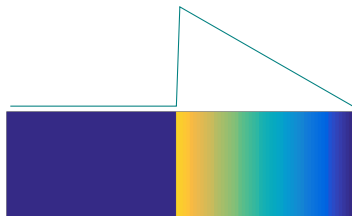


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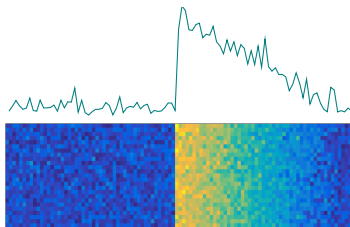


An Academic Example



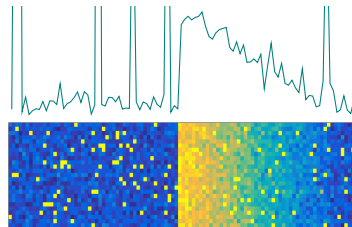
true data

- ▶ 100×25 pixels
- ▶ 18° grain boundary



Gaussian type noise

- ▶ de la Vallee Poussin
- ▶ halfwidth 1°



salt and pepper noise

- ▶ 5% missindexed pixels
- ▶ random orientation

Sliding Window Techniques for EBSD Denoising

1. the mean filter: mean of neighbors

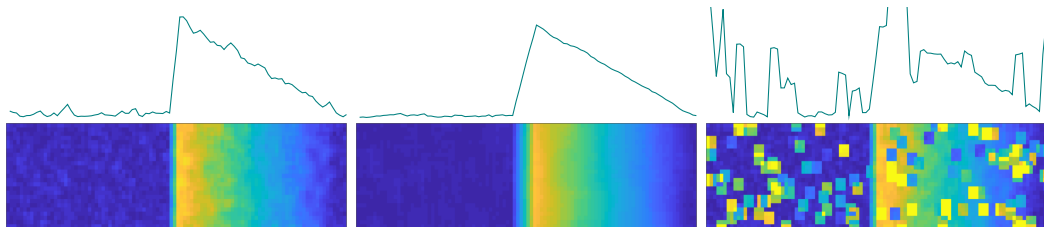
Gaussian noise ✓ grain boundaries ✗ missindexed pixels ✗ gradients ✓

2. the Kuwahara filter: mean with smallest variance

Gaussian noise ✗ grain boundaries ✓ missindexed pixels ✗ gradients ✗

3. the median filter: neighboring orientation with minimum variance

Gaussian noise ✗ grain boundaries ✓ missindexed pixels ✓ gradients ✗



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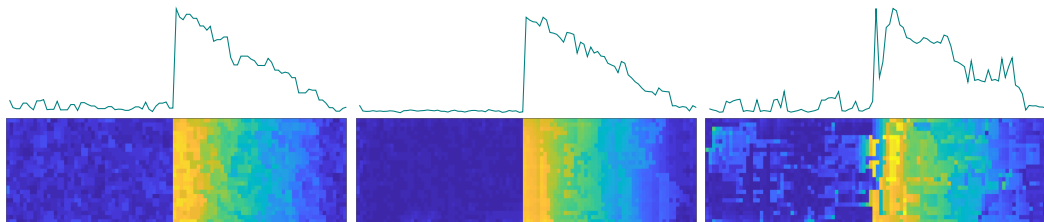
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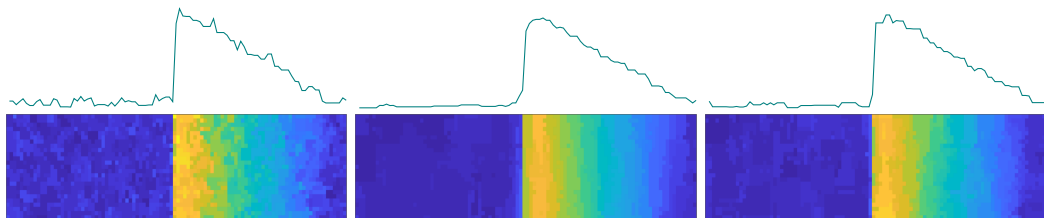
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Global Denoising Techniques

methods from mathematical image analysis

- ▶ approximation by Fourier series, splines, wavelets, shearlets
- ▶ discrete variational methods

variational methods

Minimize the functional

$$J(\vec{O}) = \sum_{i,j} \omega(\vec{O}_{ij}, \vec{O}_{ij})^2 + \alpha \cdot \varphi(\vec{O})$$

- | | |
|---|---|
| ▶ i, j - pixel indices | ▶ $\omega(\vec{O}_{ij}, \vec{O}_{ij})$ - misorientation angle |
| ▶ $\vec{O}_{ij}$ - noisy orientation image | ▶ $\varphi(\vec{O})$ - smoothness of $\vec{O}$ |
| ▶ $\vec{O}_{ij}$ - denoised orientation image | ▶ α - regularization parameter |

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- ▶ α - regularization parameter

Smoothing Splines

minimization functional

$$J(\vec{O}) = \sum_{i,j} \omega(\vec{O}_{ij}, \vec{O}_{ij})^2 + \alpha(\Delta \vec{O}_{ij})^2$$

✓ Gaussian Noise

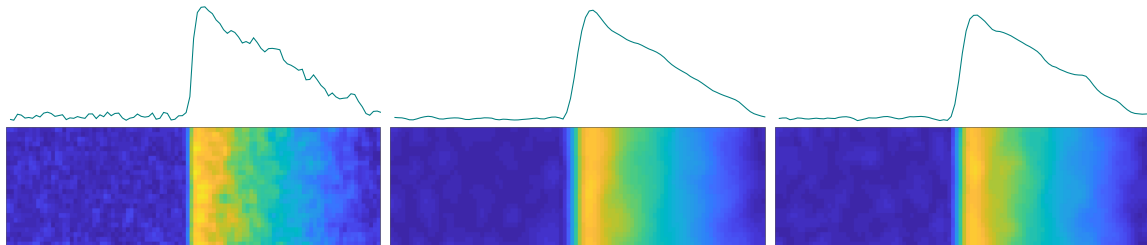
✓ gradients

✗ grain boundaries

✓ fast

✓ missindexed pixels

✓ α determination



Total Variation

minimization functional

$$J(\vec{O}) = \sum_{i,j} \omega(\vec{O}_{ij}, \vec{O})^1 + \alpha \|\nabla \vec{O}_{ij}\|_2^1$$

✓ Gaussian Noise

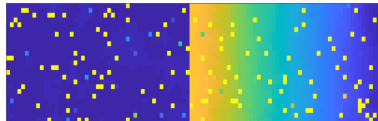
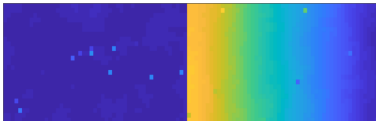
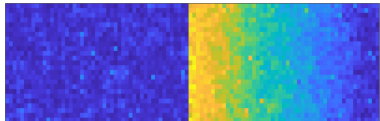
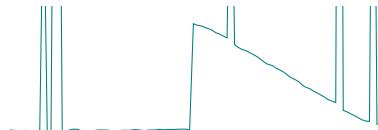
● gradients

✓ grain boundaries

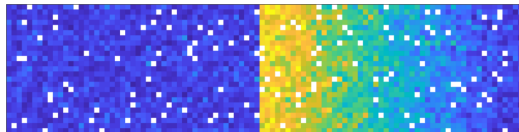
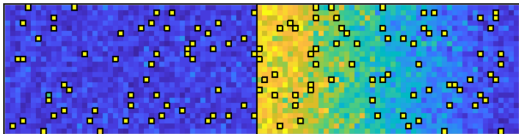
✓ currently fastest

✓ missindexed pixels

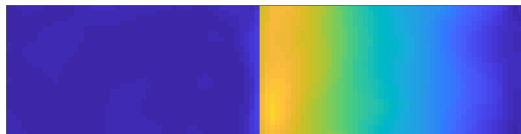
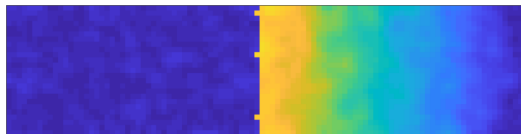
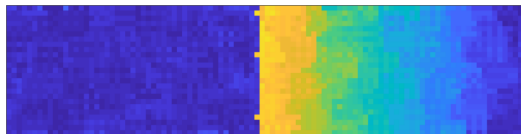
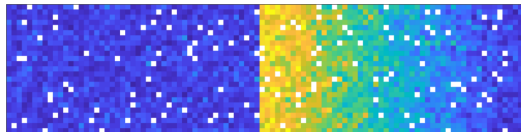
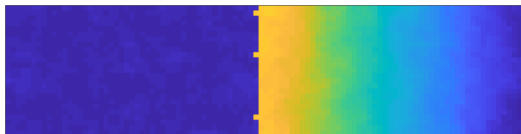
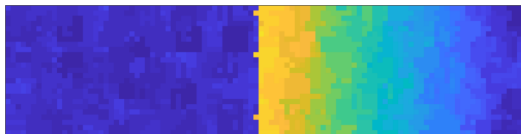
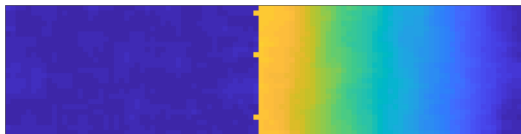
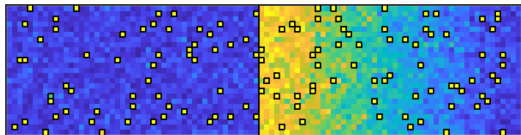
✗ α required



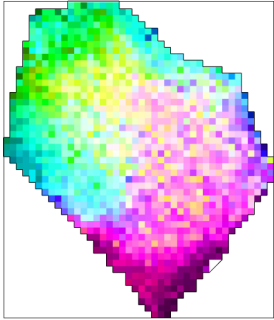
Morphological Filter



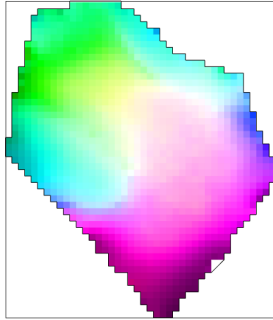
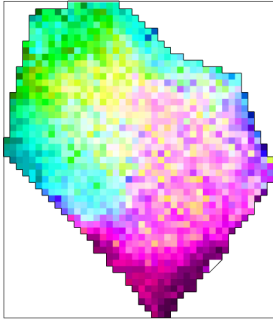
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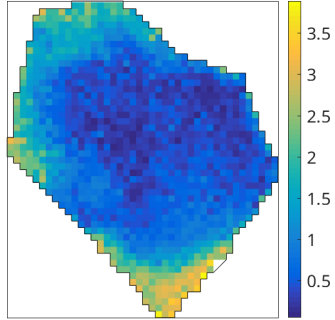
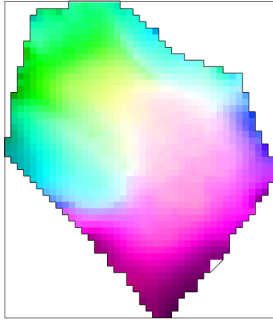
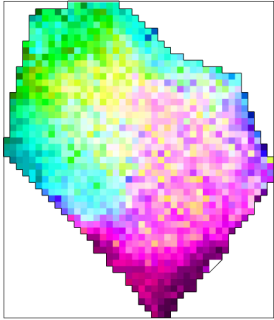
A deformed grain



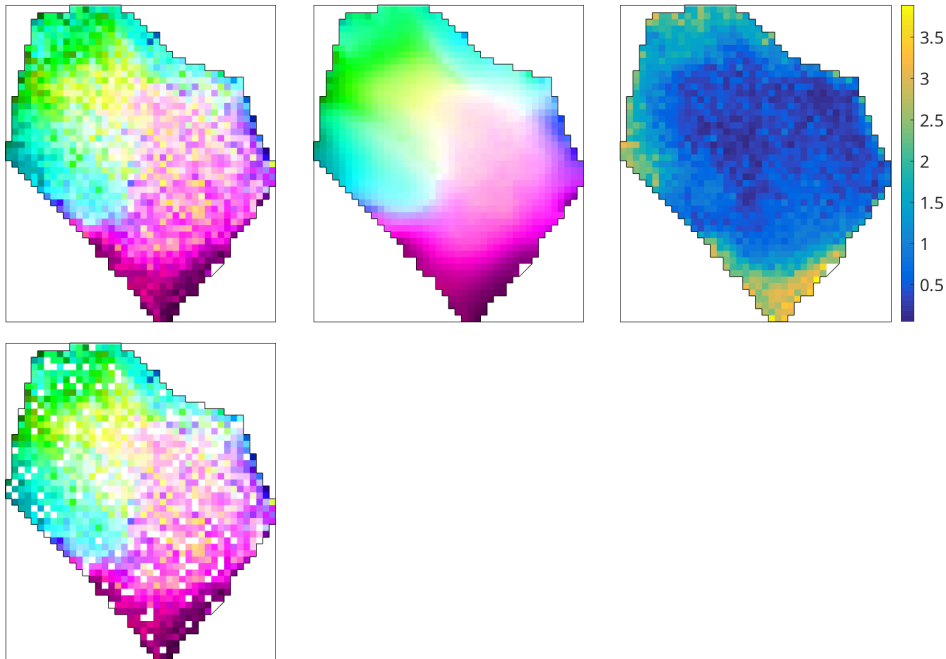
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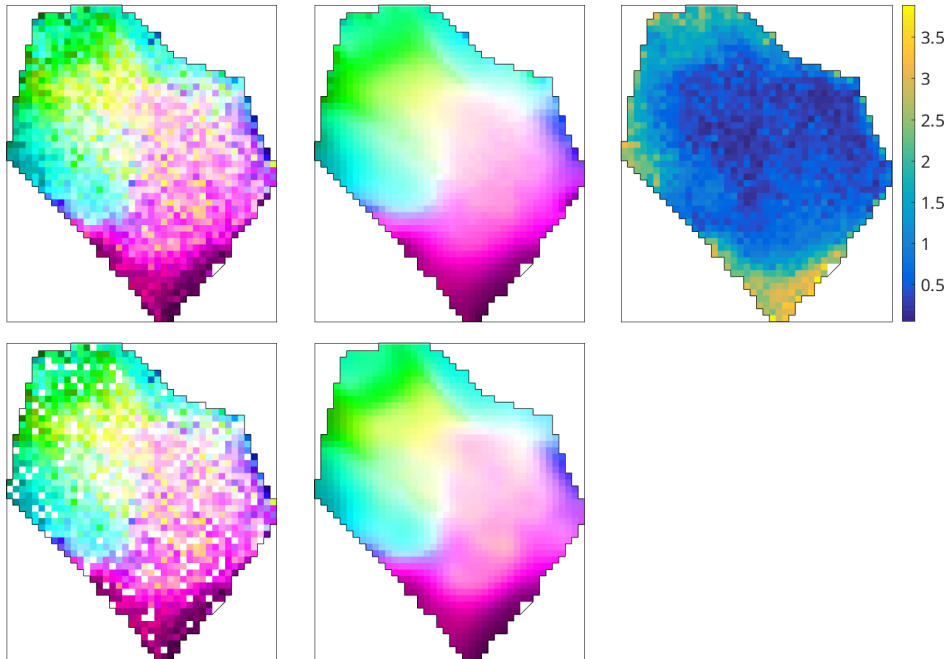
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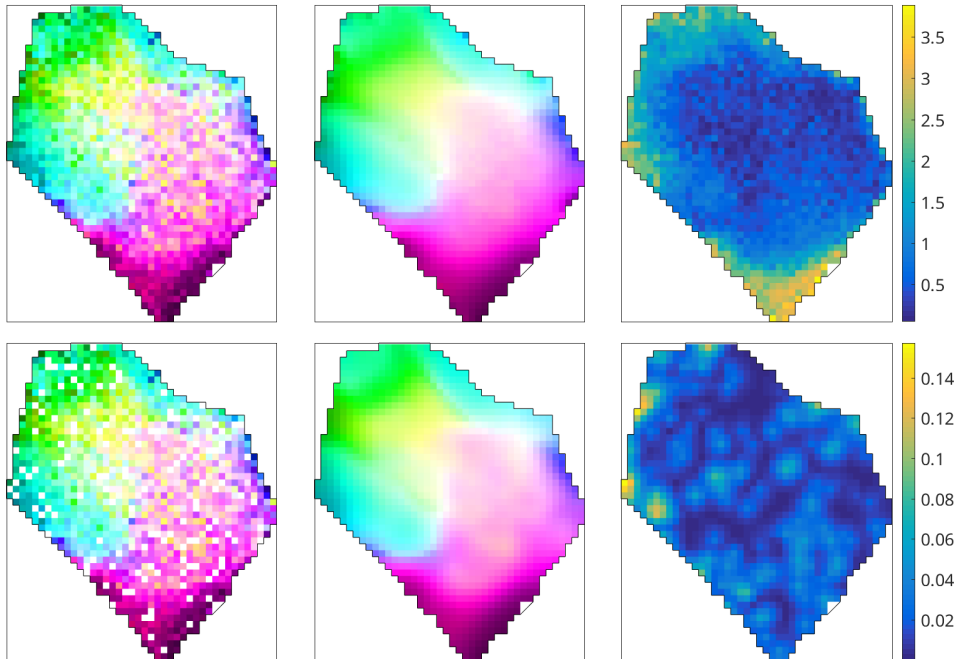
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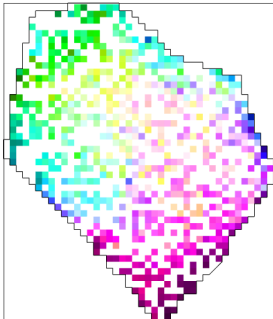
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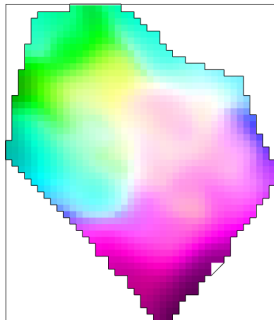
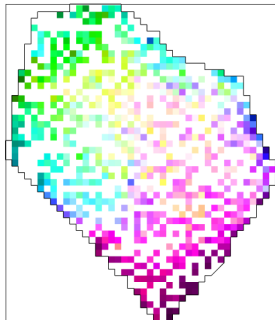
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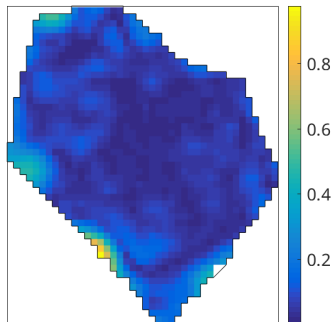
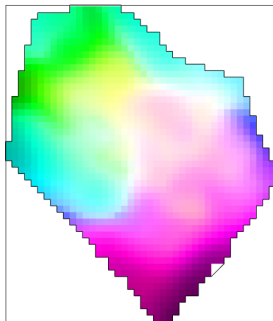
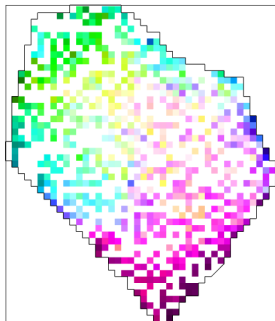
Not Indexed Data



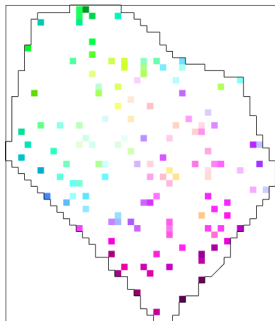
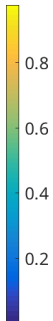
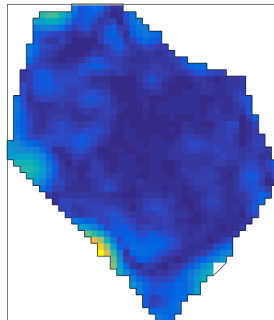
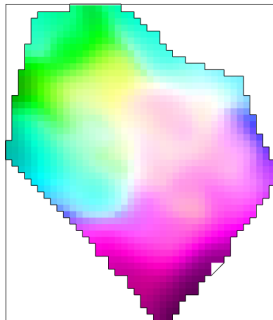
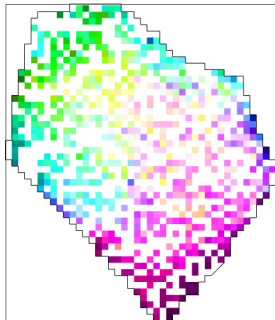
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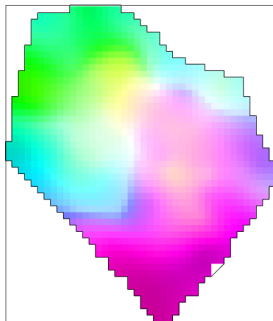
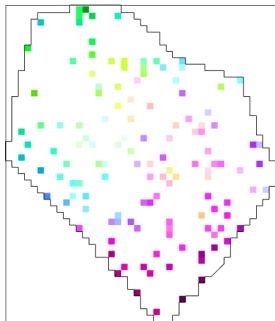
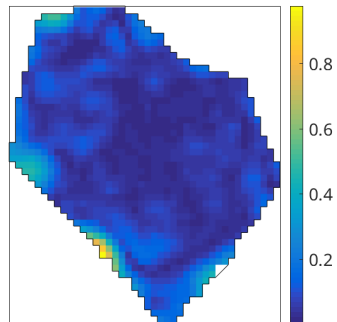
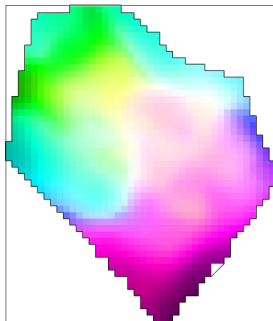
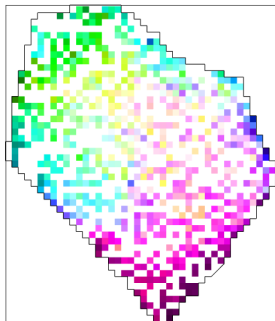
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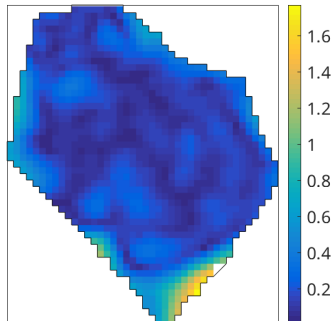
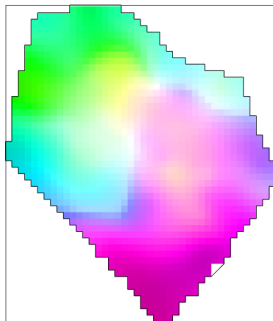
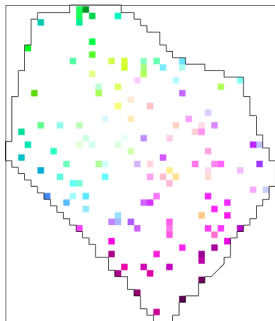
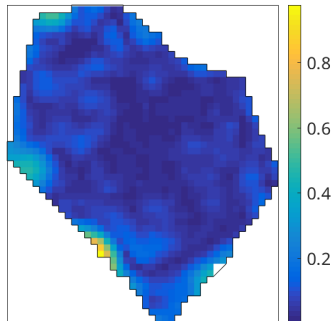
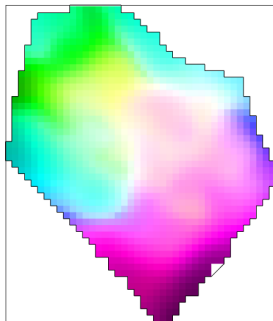
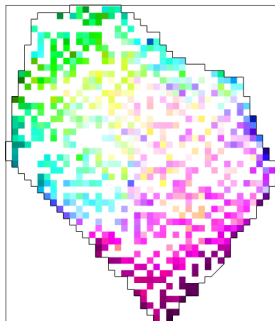
Not Indexed Data



Not Indexed Data



Not Indexed Data



0.8

0.6

0.4

0.2

1.6

1.4

1.2

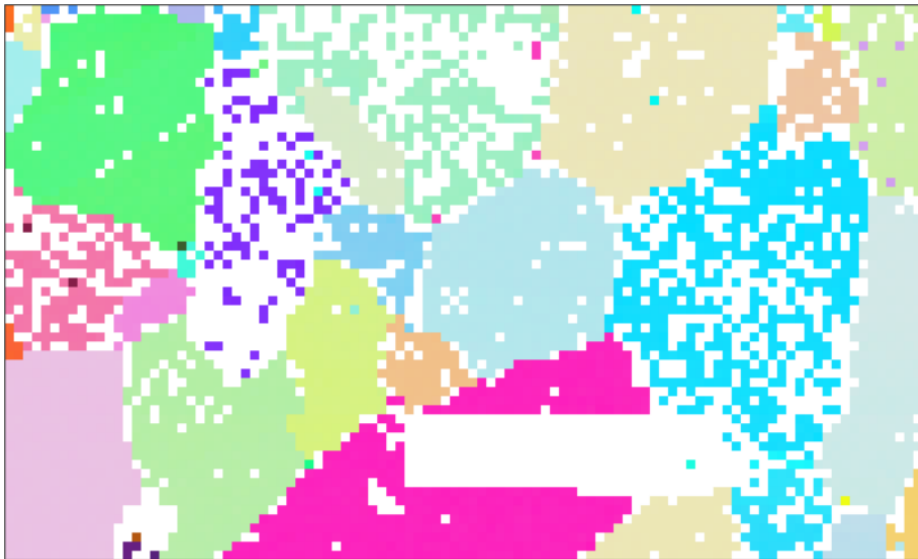
1

0.8

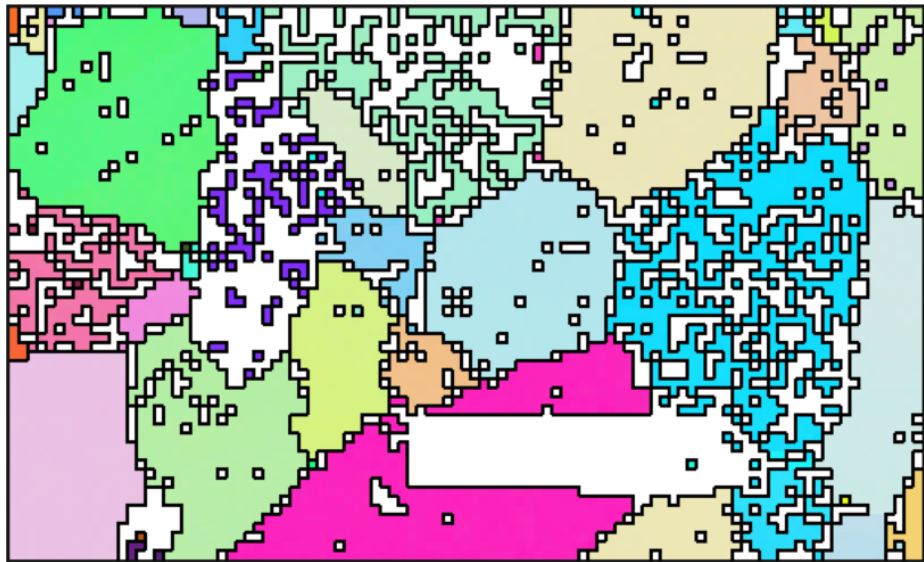
0.6

0.4

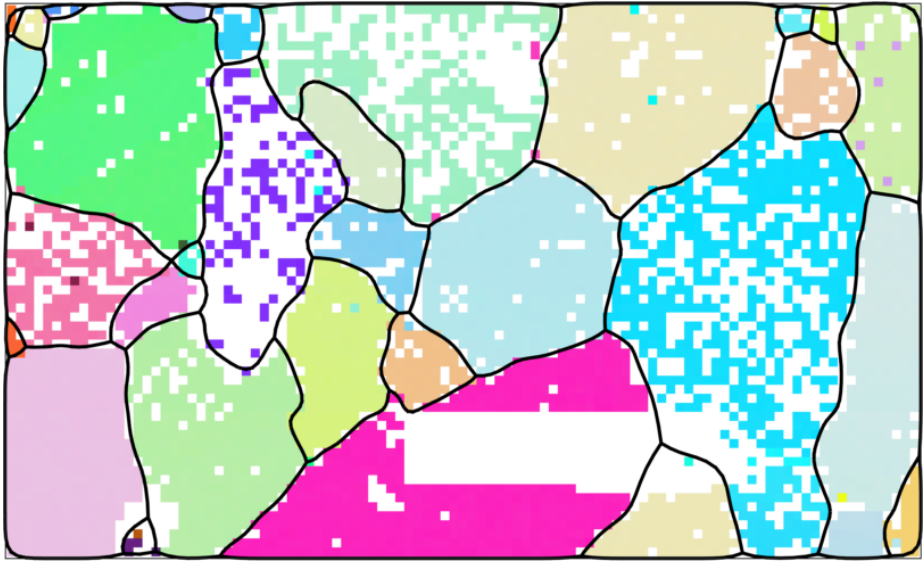
0.2



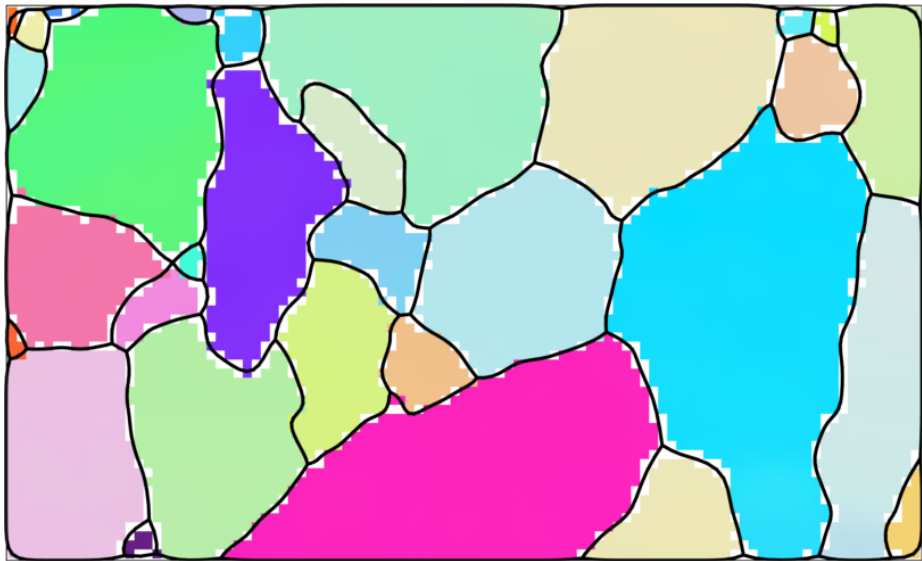
raw geological Forsterite data - white pixels are not indexed



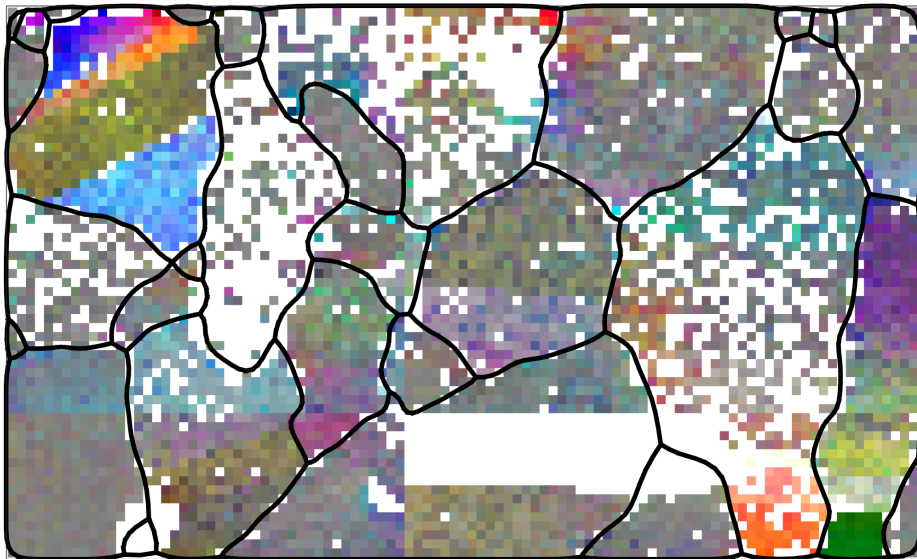
naiv grain reconstruction algorithm



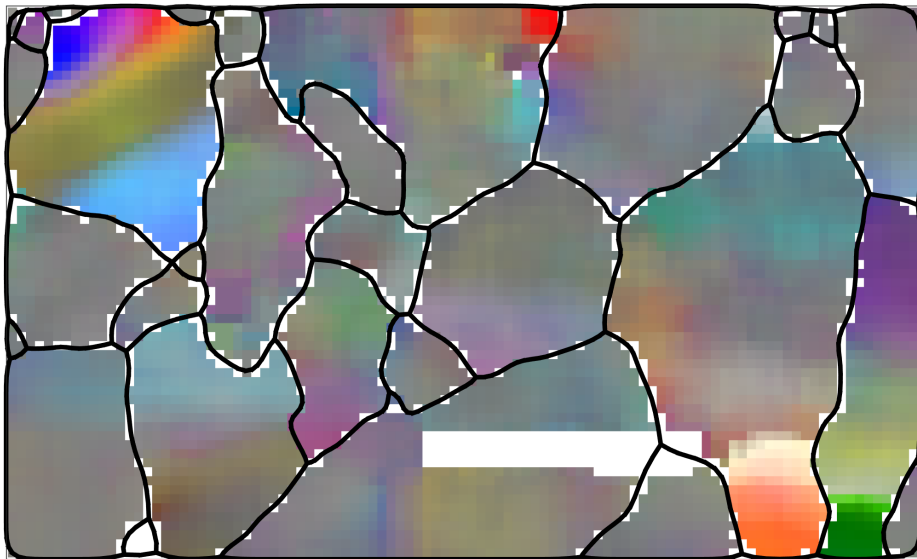
Bachmann, Hielscher, Schaeben: Grain detection from 2d and 3d EBSD data, Ultramicroscopy, 2011



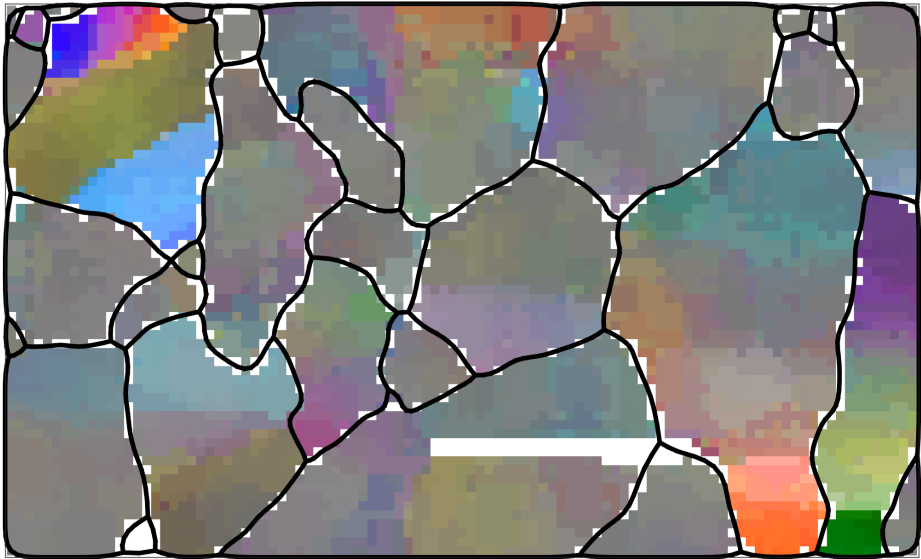
denoising and inpainting using smoothing splines



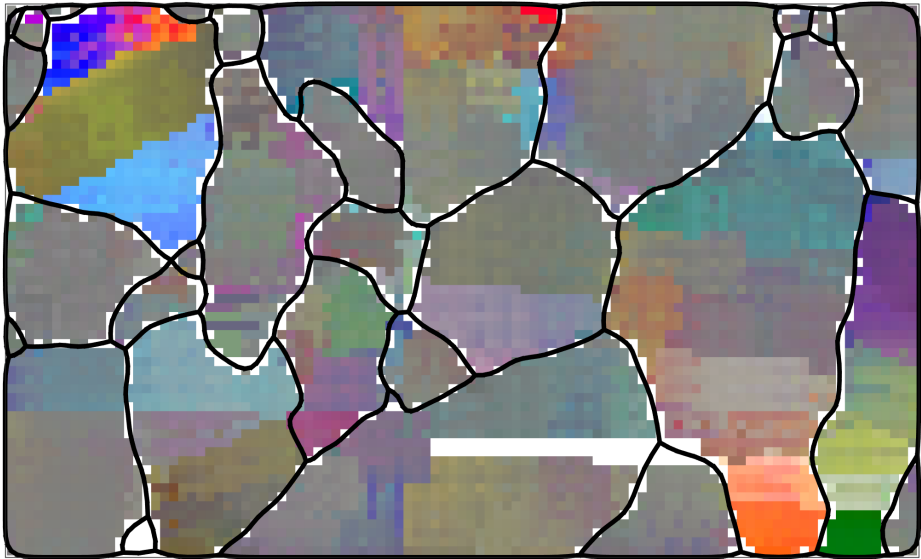
raw data, misorientation to grain mean orientation



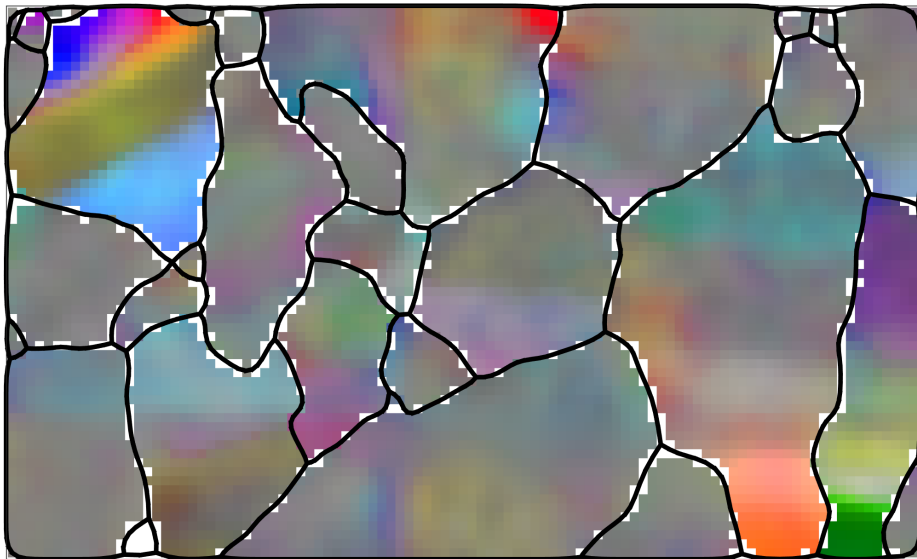
denoising and inpainting using the **mean filter** with 5×5 window



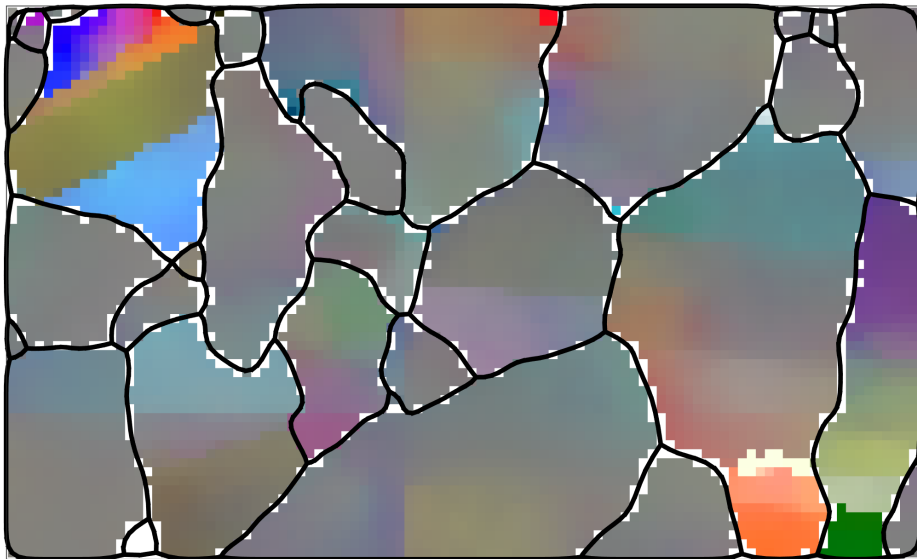
denoising and inpainting using the [median filter](#) with 7×7 window



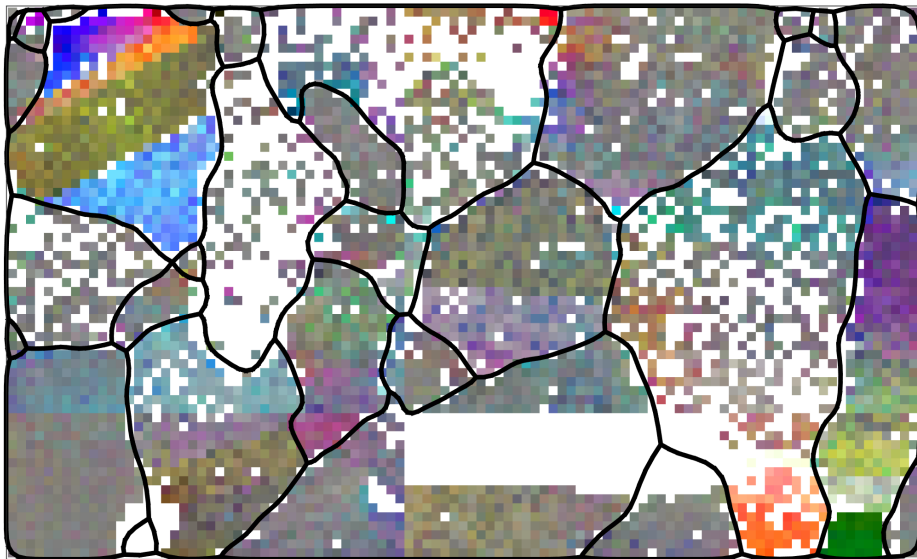
denoising and inpainting using the [Kuwahara filter](#) with 7×7 window



denoising and inpainting using [smoothing splines](#)



denoising and inpainting using [total variation](#)



raw data, misorientation to grain mean orientation

The Kernel Averaged Misorientation of Noisy Data

noise free data:



KAM:

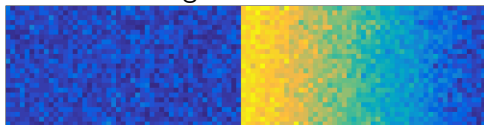


The Kernel Averaged Misorientation of Noisy Data

noise free data:



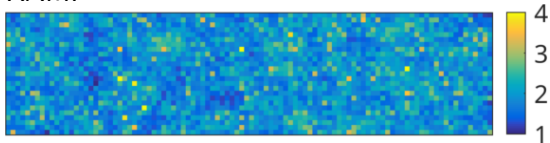
data with 1 degree noise:



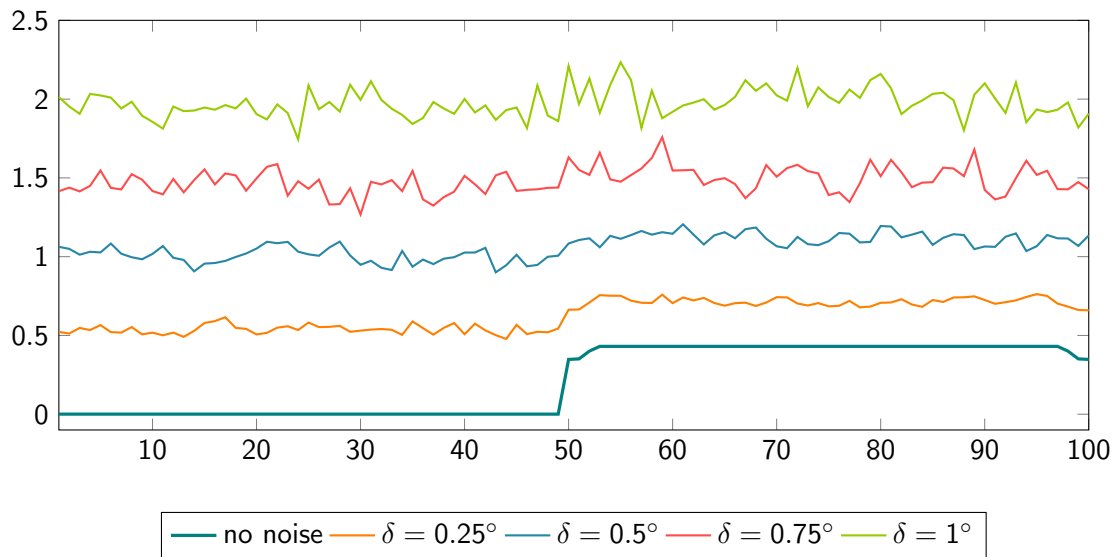
KAM:



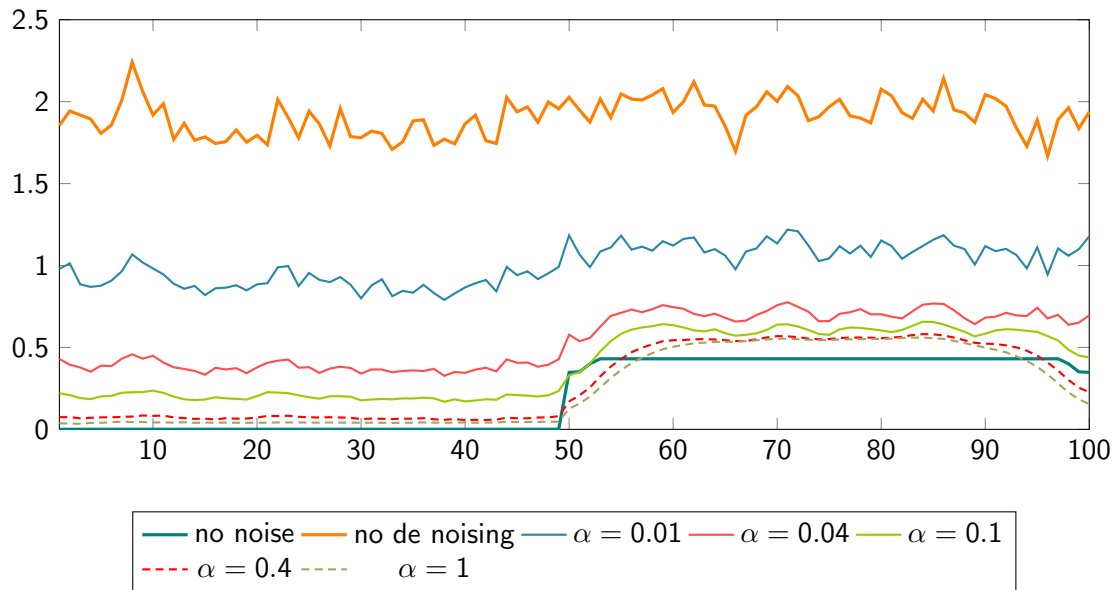
KAM:

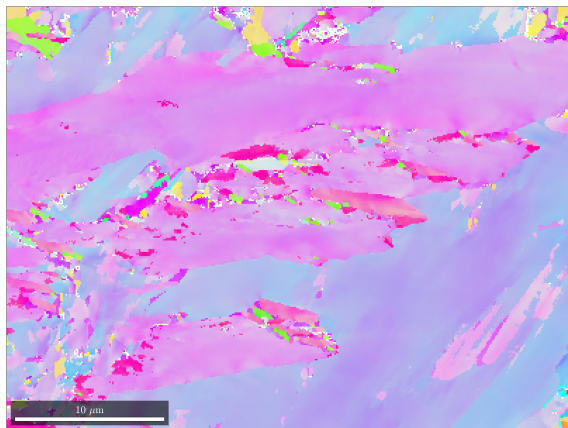
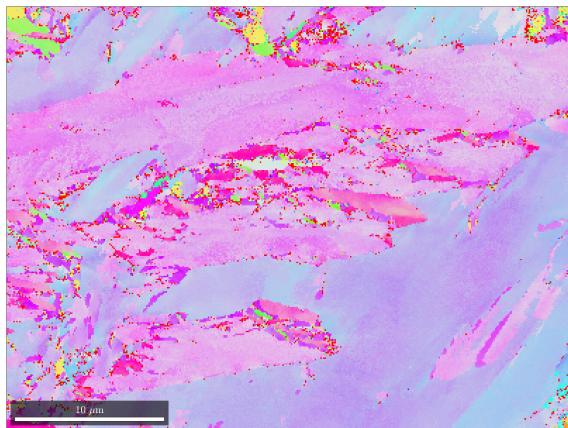


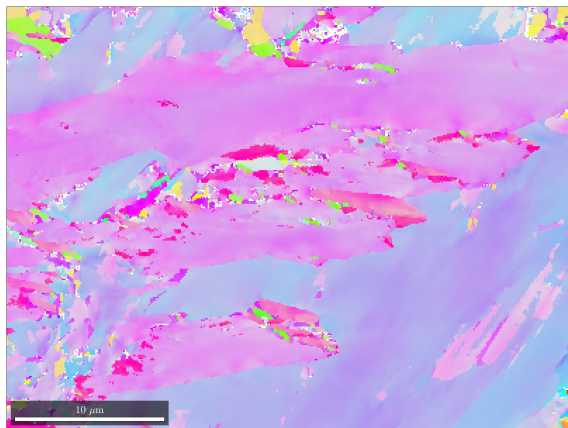
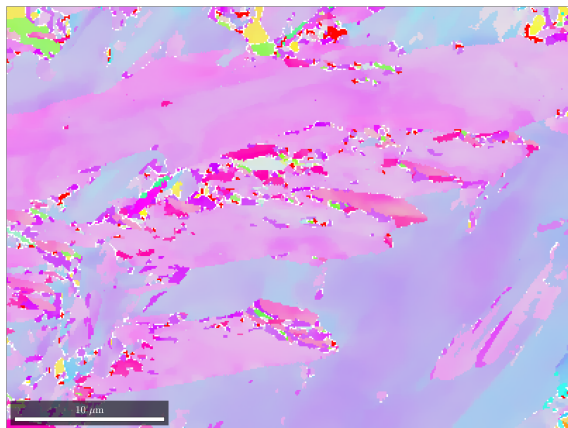
The KAM of Noisy Data



The KAM of Denoised Data







Comparison of Curvature Maps

